

Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

[Project resources](#)
[Final Report](#)

34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful

EXECUTIVE FINDING

The project confirms PRH in a narrow form: trained independent models share mainly local structure that linear maps make useful. It also bounds the claim: convergence is not global or universal, and visible geometry need not transfer.

RESEARCH PROGRAM

8-model core | 14 encoders tested | 2 shipped systems

1

Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R² mean 0.737; peak 0.797

2

Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.

R@1/5/10 = 88/95/97% native

3

Cheap Adapter hub

Seven build it; one joins later

DINOv2 s/b/l, GPT-2, BERT, SBERT and bge-m3 build the 512-d Hub and train the reusable head.

ConvNeXt: 96.7%; Hub/head stay frozen

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

1	Shared neighbours	mutual k-NN	strong locally
2	Shared order	distance-rank rho	0.29 cross-modal
3	Shared structure	CKA / RSA	partial
4	Shared subspace	ridge + retrieval	88-97% of ceiling
5	Same geometry	rotation only	insufficient: R ² = 0.038

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.592

MobileCLIP-to-SigLIP adapter R²

0.771

bge-SBERT shape rho; all 9,533 items

96.7%

new encoder; no Hub/head retraining

42 pairs

Hub route compared with direct ridge

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. Cheap Adapter tests the narrower mechanism: learned local structure is linearly usable across modalities and lineages, with held-out ConvNeXt at 96.7% of native. Limits: no universal or global identity; scaling is not monotonic beyond the tested range; the source-rank cliff is numerical.

From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

PAIRWISE

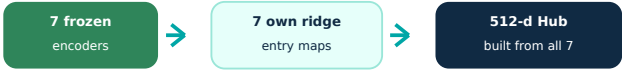
7 encoders -> 42 ordered maps

Adding one model requires maps to and from every existing model.

CHEAP ADAPTER HUB

7 encoders build Hub + get 7 maps

BUILD ONCE: the same 7 encoders define the Hub and get entry maps



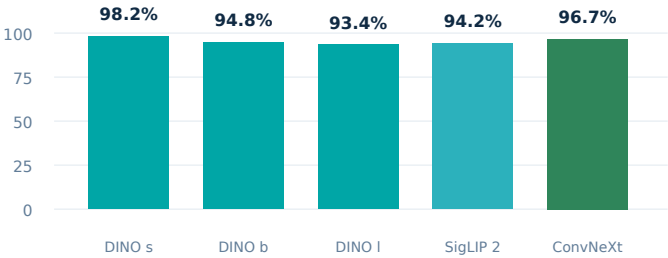
TRAIN HEAD ONCE: DINOv2-small images -> BGE-m3 caption retrieval



LATER: ConvNeXt gets one new entry map; Hub and head stay frozen -> 96.7% of native.

EACH NAMED ENCODER IS THE SOURCE: ENTRY MAP -> HUB -> FROZEN HEAD -> BGE-M3 RETRIEVAL

% of that source encoder's native BGE-m3 R@1 (1,000 held-out items)



WHAT TRANSFERRED

- The same seven encoders both build the 512-d Hub and receive their own entry maps.
- The 512 -> 1024 head trains once on DINOv2-small Hub coordinates, then freezes.
- Add ConvNeXt later by fitting only its entry map - no Hub/head retraining. It reaches 96.7% of native R@1.

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.771.

HUB / BESPOKE DIRECT

103.6%

42 ordered pairs (7 x 6). Bespoke baseline: pair-specific ridge A -> B target. Hub: A -> its map -> Hub -> frozen head -> BGE. Mean [Hub R@1 / bespoke R@1] = 103.6%; 5,376 -> 512 whitened PCA can denoise.

LIVE DEMO

39 ms / 0.08 ms

Phone indexing / local search. One query ranks phone and server items together. Adapter: 0.79 MB; 95.3% native R@5.

WHY THE INFERENCE IS CREDIBLE

Frozen models

only closed-form maps are fitted

Held-out scoring

8,533 train / 1,000 test in the hub

Negative controls

random-map R@1 = 0.001 (chance)

Nonlinear control

MLP ties linear at 0.9183

PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local - nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

Fractal probe: why shared reality matters

FractalDB tests a key PRH boundary: what structured training contributes without natural photographs, compared with matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Training creates geometry

Fractal training produces small but real shared geometry above matched random twins. Whitening reveals masked CNN structure, and the CNN signal survives PC1 removal.

G10

Geometry is not enough

That visible geometry does not become usable semantic retrieval through the frozen Hub: fractal models remain at the random floor, while ImageNet twins transfer strongly.

G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN: 0.067 fractal vs 0.036 random. Normalization explains about 10% of the gap; about 90% is training-specific.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

ENCODER	TRANSFER	NATIVE	RANDOM TWIN	INTERPRETATION
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	-	clear functional transfer
ImageNet ViT	0.204	0.208	-	clear functional transfer

PROJECT VERDICT LEDGER - 34 PRE-REGISTERED PREDICTION UNITS

21

CONFIRMED

Includes learned shared geometry, cross-modal and cross-lineage transfer, and fractal geometry above random.

7

FALSIFIED

Includes the stronger claim that visible fractal geometry must yield semantic transfer; it did not.

4

INCONCLUSIVE

Whether natural-world dependence differs systematically between image and text encoders.

2

NOT MEANINGFUL

Fractal percentage-of-native ratios, because both transfer scores lie below the 0.050 floor.

WHAT THE FRACTAL CONTROL ADDS TO PRH

Fractal training creates shared geometry above random, so convergence is learned rather than purely architectural. But retrieval remains at the random floor while ImageNet twins transfer strongly: shared natural-world experience is load-bearing, and visible geometry alone is not enough for recoverable semantic alignment.